

THIS IS TECHNOLOGY, CULTURE AND SCIENCE WITH A DIFFERENCE, AND THAT DIFFERENCE IS USUALLY ENOUGH TO MAKE MOST NORMAL PEOPLE GO "WTF?"

This month in ALT:

The Reimann hypothesis and other beautiful equations, POCO's plans for the future, and memes by interns.



The inner workings of Apple

If the Epic trial had one good fallout, it is the cache of over 5 gigabytes of internal memos of the intensely secretive company. This compilation has all the important information in a handy list, with the memos categorised according to topics such as gaming, apps and double standards.

<https://dgit.in/cookjobs>

WHAT'S NEW

Gift card gang

This is a shady network of cybercriminals which manages to compromise over 100,000 email accounts every day. Researchers suspect that the group operates by using automation to try logging in using credential leaks and dumps, using the IMAP protocol. Once the accounts are compromised, instead of using the compromised accounts for spear phishing attacks, or blasting out spam email, the cybercriminals maintain a low profile, and literally siphon out cash from the inbox. The thieves use automation to search the emails for gift cards, that to gift cards form specific sources, such as Amazon, Starbucks, Microsoft and PayPal. The automation scans for



gift cards from crypto service providers as well. These gift cards are harvested and then sold to third parties over sidey

internet marketplaces, where they fetch around 80 percent of the original value. Now the thieves are so aggressive about obtaining the gift cards, that they also complete signups, opt in for offers, and go through the hoops of the series of clicks necessary to complete reward programs, and get even more gift cards, or cash credited in gift cards all automated on the "customer's" behalf. <https://dgit.in/gcg>

Smart dental implants

Dental implants often need to be replaced within five to ten years because of inflammation or gum disease, well short of their intended use for twenty years. Researchers from the University of Pennsylvania have developed smart dental implants that attack the problem in two ways, the first is a nanoparticle infused material that prevents bacterial colonies from cropping up. The material is also



piezoelectric, meaning that it can generate electricity from the natural motions of the mouth such as chewing or brushing. The energy is used to power up a series of lights

embedded within the gums, that provide phototherapy to the gums, to curb the growth of pathogens and encourage tissue growth. The implants have demonstrated their safety and longevity in tests, and now the researchers are moving the commercialise the product. What this means is that it won't be long before some biohacker comes up with RGB dental implants. <https://dgit.in/sdi>



Insta Kids paused

Instagram announced in a blog post that it would be pausing development of a kids version of the platform with age appropriate features such as a lack of targeted ads, parental supervision tools, and prompts to take a break if they are spending too much time scrolling. <https://dgit.in/instakids>



Personality tests

A close look at the MBTI personality tests, how it became so popular, and the entities that benefit from the "typing". Known as "astrology for businessmen", the tests sell people their own personalities as commodities. There is a still darker side to the tendency of sorting people. <https://dgit.in/mbti>



Engineering caustics

You know the weird patterns that are made by variations in the surface of glass? This DIY tutorial shows you how to hack a piece of acrylic, to form caustics from any image. The "magic windows" project an image when light is shone through them, and show caustics in the sun. <https://dgit.in/causkits>